Identifiability Analysis of a Multi-Cell Line CHO Kinetic Model Using a Python Framework

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***Abstract*—The Multi-Cell Line Kinetic Model (MCKM) of CHO fed-batch cultures was implemented in Python, with explicit bolus feeding, to assess the identifiability of its 13 kinetic parameters. Local structural identifiability was evaluated by Singular Value Decomposition of the Relative Output Sensitivity Matrix (SVD-ROSM), and its robustness by Monte Carlo sampling of 100 parameter sets (±20%). The full model showed a smallest singular value $\sigma_{13} \approx 1 \times 10^{-15}$, caused by an exact dependency between $Y_{X/gln}$ and $Y_{gln/glu}$. Fixing $Y_{gln/glu}$ restored structural identifiability ($\sigma_{12} \approx 5 \times 10^{-3}$). Under 5% measurement noise, however, t-value analysis showed that K_{glc}, $Y_{X/lac}$ and m_{glc} were practically non-identifiable; fixing them yielded a 9-parameter model with $|t| \geq 2.03$ for all free parameters. Unlike the pairwise collinearity index (CI) as applied in previous work, SVD-ROSM recovers the non-identifiable parameter combination in a single decomposition and distinguishes genuine structural dependencies from apparent ones. The open-source framework supports reproducible identifiability assessment and data-driven clone selection in CHO cell line development.**

Keywords—CHO cells, structural identifiability, practical identifiability, sensitivity analysis, singular value decomposition, Monte Carlo analysis

I. INTRODUCTION

The production of monoclonal antibodies (mAbs) is one of the fastest-growing therapeutic areas in the biopharmaceutical industry, accounting for 63% of a USD 484 billion global market in 2025 [1], [2]. Chinese Hamster Ovary (CHO) cells are the dominant production platform owing to their scalability, human-like post-translational modifications, and adaptability to diverse protein products [2]–[4]. However, the development of clonal CHO cell lines (CLD) is challenged by their intrinsic genomic plasticity, which leads to high clonal heterogeneity and variable metabolic behavior across clones [4]–[6]. Kinetic modeling has become a fundamental tool for analyzing and predicting CHO cell growth, metabolism, and protein production, supporting process optimization, scale-up, and rational cell line engineering [4]–[6].

Traditional kinetic approaches require multiple culture runs to calibrate parameters, which increases the experimental workload and limits their applicability across diverse clones [3]. The Multi-Cell Line Kinetic Model (MCKM) addresses these limitations by deriving a complete set of 13 kinetic parameters from a single culture run with only 49 data points, enabling the in silico simulation and metabolic characterization of individual clones [3]. However, the limited data relative to the size of the parameter space introduces risks of overfitting and parameter correlation, so identifiability must be assessed to ensure that parameter differences reflect biological variation rather than mathematical artifacts [3], [5], [7].

Structural identifiability evaluates whether model parameters can be uniquely inferred from perfect, noise-free data [7]. If a model is not structurally identifiable, unique parameter values cannot be assigned regardless of data quality or estimation method, and apparent differences between clones may arise from mathematical correlations rather than from biology [3], [7]. Practical identifiability additionally considers experimental noise and sparse measurements [7], [8], ensuring that parameters remain meaningful under realistic conditions.

In [3], identifiability was assessed with a collinearity index (CI) computed pairwise on time-averaged sensitivities in MATLAB. The CI can be extended to larger parameter groups through combinatorial search [9], but it quantifies the degree of collinearity of a given subset rather than the non-identifiable parameter combination itself. Singular Value Decomposition (SVD) of the Relative Output Sensitivity Matrix (ROSM), proposed by Stigter and Molenaar [7], determines this combination directly: it identifies complete linear combinations of non-identifiable parameters in a single decomposition, scales efficiently to large models, and is readily implemented in open-source Python tools [8].

This study aims to evaluate the structural and practical identifiability of the MCKM using an SVD-ROSM framework in Python. The objectives are: (1) to implement the MCKM ODEs with fed-batch dynamics; (2) to perform local structural identifiability and Monte Carlo robustness analyses; (3) to fix non-identifiable parameters to restore identifiability; (4) to examine practical identifiability through t-value analysis under experimental noise; and (5) to compare the results with those of Sietaram et al. [3] and with other CI-based approaches [9].

Section II presents the model formulation, its numerical implementation, and the structural and practical identifiability methods. Section III presents the simulation verification, the structural and practical identifiability results, and the comparison with CI-based approaches. Section IV summarizes the conclusions and their implications for clonal characterization.

II. METHODOLOGY

A. Dynamic Model Description

To address clone variability in biopharmaceutical production, this study adopts the MCKM proposed by Sietaram et al. [3], a generalized mechanistic model that extracts a complete set of kinetic parameters from a single fed-batch culture run. The nonlinear state-space model consists of eight coupled ordinary differential equations (ODEs) describing viable biomass (X_v), monoclonal antibody titer (P), five key metabolites (glucose, glutamate, glutamine, ammonium, and lactate), and an auxiliary state for the specific growth rate (μ).

Modeling assumptions:

The model follows macroscopic, unsegregated, and unstructured bioprocess modeling principles. The bioreactor is assumed to be perfectly mixed, so temperature, pH, and concentrations are spatially homogeneous. Growth is limited only by glucose, the primary carbon source, whereas glutamate is fed in excess and is not growth-limiting. Cell death is driven exclusively by the accumulation of cytotoxic ammonium. The culture operates in fed-batch mode with variable volume and discrete bolus feeding of glucose and glutamate over 15 days.

Dynamic mass balances:

Growth follows Monod kinetics limited by glucose, whereas by-product formation and mAb production follow Luedeking–Piret relationships. A central feature of the model is a mechanistic lactate switch, which governs the transition from lactate production to consumption when glucose falls below a threshold α_{glc}. The governing equations are [3]:

Cellular growth and death:

$$\frac{dX_v}{dt} = \big(\mu(t) - \mu_d(t)\big) \cdot X_v(t) \tag{1}$$

$$\mu(t) = \frac{\mu_{max} \cdot glc(t)}{K_{glc} + glc(t)} \tag{2}$$

$$q_{glu}(t) = \frac{\mu(t)}{Y_{X/glu}} \tag{3}$$

$$\mu_d(t) = k_D \cdot \frac{amm(t)}{KD_{amm} + amm(t)} \tag{3}$$

Product formation and substrate consumption:

$$\frac{dP}{dt} = q_p(t) \cdot X_v \tag{4}$$

$$q_p(t) = \mu(t) \cdot Y_{\frac{P}{X}} \tag{5}$$

$$\frac{dglc}{dt} = -q_{glc}(t) \cdot X_v(t) + F(t) \cdot \frac{glc_{F,in} - glc(t)}{V_L(t)} \tag{7}$$

$$\frac{dglu}{dt} = -q_{glu}(t) \cdot X_v(t) + F(t) \cdot \frac{glu_{F,in} - glu(t)}{V_L(t)} \tag{6}$$

$$q_{glc}(t) = \frac{\mu(t)}{Y_{X/glc}} + m_{glc} \tag{7}$$

$$\frac{dgln}{dt} = -q_{gln}(t) \cdot X_v(t) + Y_{\frac{gln}{glu}} \cdot q_{glu}(t) \cdot X_v(t) \tag{8}$$

$$q_{gln}(t) = \frac{\mu(t)}{Y_{X/gln}} \tag{9}$$

$$\frac{damm}{dt} = q_{amm}(t) \cdot X_v(t) \tag{10}$$

$$q_{amm}(t) = \frac{\mu(t)}{Y_{X/amm}} \tag{11}$$

Metabolic switch (lactate):

$$\frac{dlac}{dt} = q_{lac}(t) \cdot X_v(t) \tag{12}$$

$$if\ [glc] > \alpha_{glc}; q_{lac}(t) = Y_{\frac{lac}{glc}} \cdot q_{glc}(t); \tag{13}$$

$$else: q_{lac}(t) = -\frac{\mu(t)}{Y_{\frac{X}{lac}}} \cdot X_v(t) + Y_{\frac{lac}{glc}} \cdot q_{glc}(t) \tag{14}$$

Auxiliary growth-rate constraint:

To stabilize the numerical regression, the derivative of the growth rate is included as an additional state, obtained by differentiating (2) [3]:

$$\frac{d\mu}{dt} = \mu_{max} \frac{\frac{dglc}{dt} \cdot K_{glc}}{\left(K_{glc} + glc\right)^2} \tag{15}$$

The 13 kinetic parameters θ define the metabolic profile of each clonal cell line; their nominal values (Table I) lie within the ranges reported in [3].

TABLE I. KINETIC PARAMETERS AND NOMINAL VALUES

Symbol	Description	Units	Value
μ_{max}	Maximum growth rate	1/h	0.0257
k_d	Death rate constant	1/h	0.0160
K_{glc}	Glucose saturation constant	mM	1.0500
$Y_{\frac{X}{glc}}$	Biomass yield on glucose	10^9 cells/mM	2.0247
$Y_{\frac{X}{glu}}$	Biomass yield on glutamate	10^9 cells/mM	8.9000
$Y_{\frac{X}{gln}}$	Biomass yield on glutamine	10^9 cells/mM	43.2000
$Y_{\frac{lac}{glc}}$	Lactate yield on glucose	mol/mol	0.2800
$Y_{\frac{X}{amm}}$	Biomass yield on ammonium	10^9 cells/mM	11.000
$Y_{\frac{gln}{glu}}$	Glutamine yield on glutamate	mol/mol	0.3800
$Y_{\frac{X}{lac}}$	Biomass yield on lactate	10^9 cells/mM	3.1500
$Y_{\frac{P}{X}}$	mAb yield on biomass	mmol/10^9 cells	0.0450
m_{glc}	Glucose maintenance coefficient	mmol/(L·h)	3.15e-6
KD_{amm}	Ammonium inhibition constant	mM	6.2000

B. Numerical Implementation

To establish a robust basis for the identifiability analysis, the MCKM was implemented in Python. The coupling between continuous metabolic dynamics and discrete bolus feeding makes the system numerically stiff, so it was simulated with CasADi, an algorithmic differentiation and optimization framework, using the implicit IDAS solver. Its backward differentiation formulas (BDF) handle the feeding discontinuities robustly under strict tolerances (atol = 10^{-8}, rtol = 10^{-6}). The initial conditions are listed in Table II.

TABLE II. INITIAL CONDITIONS USED IN THE SIMULATIONS

X_v (10^9 cells)	P (mM)	Glc (mM)	Glu (mM)	Gln (mM)	Amm (mM)	Lac (mM)	μ (h^{-1})
0.3	0.0	20.0	8.0	2.0	0.2	1.0	0.024

All values were fixed except μ_0, which was computed from (2) with the initial glucose concentration.

C. Structural Identifiability Analysis

A model is structurally identifiable if, under perfect, noise-free data, its parameters θ can be uniquely determined; otherwise, observed differences between clones could be artifacts of mathematical correlation rather than true biological divergence [7]. Following Stigter and Molenaar [7], the analysis is based on the parametric sensitivities of the outputs y_i with respect to the parameters θ_j:

$$S_{ij}(t) = \frac{\partial y_i(t)}{\partial \theta_j} \tag{16}$$

To prevent scaling distortions arising from disparate parameter and output magnitudes, the sensitivities are normalized into the Relative Output Sensitivity Matrix (ROSM):

$$\tilde{S}_{i,j}(t) = \frac{\theta_j}{y_i} \frac{\partial y_i}{\partial \theta_j} \tag{17}$$

SVD is then applied to the ROSM to assess its rank deficiency:

$$\tilde{S} = U \Sigma V^T \tag{18}$$

The diagonal matrix Σ contains the singular values σ_k. Singular values close to zero (below the numerical threshold of 10^{-4}) indicate rank deficiency, and the corresponding right singular vectors in V define the linear combinations of parameters that cannot be decoupled even with ideal data. The parameters dominating these directions are flagged as structurally non-identifiable and fixed at biologically plausible values, yielding a reduced model suitable for clone screening.

To reflect the experimental constraints of the CLD data in [3], sensitivities were evaluated only at the available sampling times. The culture runs provide 49 data points (7 measured outputs at 7 sampling times), so the exact Jacobian of the outputs with respect to the parameters was computed at these times by forward algorithmic differentiation in CasADi. The relative sensitivities were stacked into a 49 × 13 ROSM, in which each row corresponds to one output at one sampling time and each column to one kinetic parameter.

Because SVD-ROSM evaluates local structural identifiability, its conclusions may depend on the nominal parameter values. The analysis was therefore repeated for 100 parameter sets generated by uniform random perturbations of ±20% around the nominal values, accounting for expected biological variability and analytical uncertainty.

D. Practical Identifiability Analysis

Practical identifiability was assessed with theoretical t-values, assuming a uniform relative measurement error of 5%. The relative parameter covariance was approximated from the Fisher Information Matrix built with the reduced ROSM S_{red} (the full 13-parameter FIM is singular):

$$\mathrm{Cov}_{\mathrm{rel}} = 0.05^2 \left(\tilde{S}_{\mathrm{red}}^{\mathrm{T}} \tilde{S}_{\mathrm{red}}\right)^{-1} \tag{19}$$

The t-value of each parameter is the inverse of its relative standard error, i.e., of the square root of the corresponding diagonal element of Cov_{rel}. Parameters with $|t|$ below the two-sided 95% Student's t value for $49 - p$ degrees of freedom ($\approx$ 2.03 for p = 12) were considered practically non-identifiable. The analysis was repeated over the Monte Carlo parameter sets. Since no parameters were fitted to experimental data, these t-values quantify the expected (a priori) practical identifiability at the nominal parameter values, not the precision of actual estimates.

E. Parameter-Fixing Strategy

To mitigate both structural rank deficiencies and practical non-identifiability, parameters that dominated the null space, or frequently showed $|t| < 2.03$, were fixed iteratively at their nominal values (with recalculation between iterations) until the minimum singular value σ_{min} exceeded the numerical threshold and all remaining free parameters achieved significant t-values.

Finally, the SVD-ROSM and t-value results were compared with the pairwise CI analysis of [3] and with the CI framework of Gábor et al. [9].

III. RESULTS AND DISCUSSION

A. Simulation of the Fed-Batch MCKM Model

Using the parameters, initial conditions, and feeding schedule of [3], our simulations qualitatively reproduced the reported biomass growth, subsequent decline, and progressive mAb accumulation (Fig. 1). Only X_v and mAb are shown because they were the primary outputs used to validate the reference model, which reached $R^2 > 0.90$ after calibration [3]; the remaining metabolite profiles are available in the Supplementary Data. This agreement supports the use of these trajectories for the subsequent identifiability analyses.

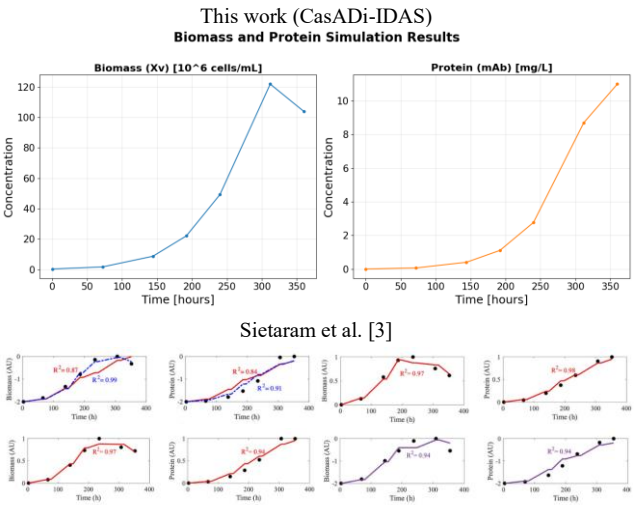


Fig. 1. MCKM simulations of biomass and mAb production. Top: this work (CasADi-IDAS); bottom: Sietaram et al. [3] (normalized units).

B. Structural Identifiability Assessment

Structural identifiability of the full 13-parameter MCKM was evaluated through SVD of the ROSM. As shown in Fig. 2, the smallest singular value was $\sigma_{13} = 1.08 \times 10^{-15}$, several orders of magnitude below the numerical threshold of 10^{-4}, confirming a rank deficiency of the full model. The corresponding null-space vector shows that this loss of rank is driven by a dependency between $Y_{X/gln}$ and $Y_{gln/glu}$, while the remaining parameters contribute only marginally to this direction. The Monte Carlo analysis with 100 perturbed parameter sets (±20%) confirmed that this result does not depend on a particular nominal parameter set but is a structural property of the model.

Unlike the constrained-regression strategy of Sietaram et al. [3], the SVD-ROSM framework bases parameter fixing on the mathematical structure of the model. In the original study, when the initial fit yielded $R^2 < 0.90$ for X_v or mAb, a new regression was automatically triggered in a 12-parameter space by fixing either μ_{max} or K_{glc} at its initial estimate. Although this strategy improved poor fits, parameter selection depended on an empirical goodness-of-fit criterion. In contrast, σ_{13} and its null-space vector show that the structural rank deficiency is not dominated by the growth parameters but by the dependency between $Y_{X/gln}$ and $Y_{gln/glu}$. The proposed approach thus replaces fit-triggered parameter selection with a reproducible, mathematically grounded model-reduction strategy.

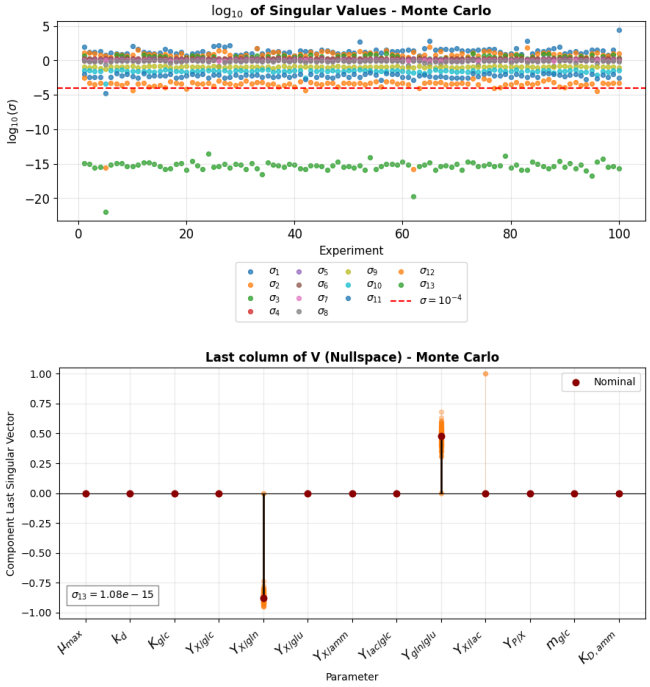


Fig. 2. Structural non-identifiability of the full MCKM: (top) singular values for 100 Monte Carlo parameter sets; (bottom) dominant null-space direction, associated with $Y_{X/gln}$ and $Y_{gln/glu}$.

C. Model Reduction

Since the dominant non-identifiable direction is associated with $Y_{X/gln}$ and $Y_{gln/glu}$, $Y_{gln/glu}$ was fixed at the nominal value reported in [3] (0.38 mol/mol), which is bounded above by the 1:1 stoichiometry of glutamine synthetase. The near-zero singular value indicates a null-space direction along which changes in both parameters compensate each other, producing virtually the same model response. Fixing $Y_{gln/glu}$ removes this redundant direction, breaks the linear dependency between the corresponding sensitivity columns, and leaves $Y_{X/gln}$ free as the clone-dependent parameter. As shown in Fig. 3, all singular values of the reduced 12-parameter model then exceed the threshold ($\sigma_{12} \approx 5 \times 10^{-3}$). This step is necessary before practical identifiability analysis, since a strong structural dependency can mask smaller but experimentally relevant parameter variations under realistic noise.

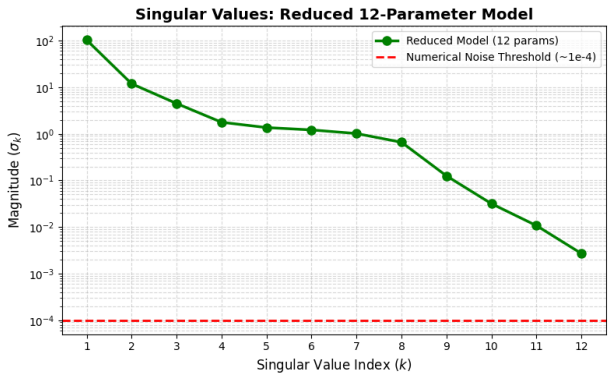


Fig. 3. Singular values of the reduced 12-parameter MCKM after fixing $Y_{\mathrm{gln/glu}}$; all remain above the numerical threshold of 10^{-4}.

D. Practical Identifiability

After fixing $Y_{\mathrm{gln/glu}}$, the reduced 12-parameter model was structurally identifiable but not practically identifiable under an assumed noise level of 5% (Fig. 4, top). K_{glc}, $Y_{\mathrm{X/lac}}$ and m_{glc} remained below the critical threshold $|t| = 2.03$, indicating that their estimates would not be sufficiently reliable under realistic experimental uncertainty.

The poor practical identifiability of K_{glc} and m_{glc} provides an additional explanation for the glucose-modeling anomaly reported by Sietaram et al. [3]. For the cell lines producing mAb-A, that study obtained an average glucose coefficient of determination of $R^2 = -0.24$, which was mainly attributed to approximate feeding schedules and to the sensitivity of R^2 to experimental deviations. The present t-value analysis indicates that the glucose affinity and maintenance parameters are also weakly estimable under measurement noise. Therefore, the poor predictive performance for glucose may reflect not only operational uncertainty but also insufficient information in the data to distinguish the individual effects of K_{glc} and m_{glc}.

Systematically fixing $Y_{\mathrm{gln/glu}}$, K_{glc}, $Y_{\mathrm{X/lac}}$ and m_{glc} progressively improved practical identifiability (Fig. 4, bottom), yielding a 9-parameter model with $|t| \geq 2.03$ for all free parameters. Further fixing KD_{amm} and $\mu_{\max}$, guided by their frequently low t-values across Monte Carlo simulations, reduces the model to seven free parameters.

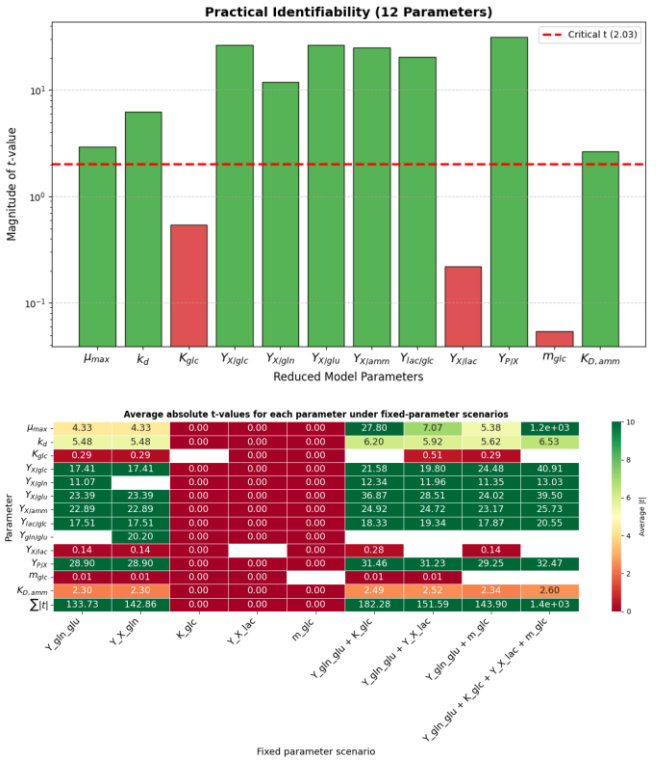


Fig. 4. Practical identifiability under 5% noise: (top) t-values of the 12-parameter model and critical value $|t| = 2.03$; (bottom) mean $|t|$ across Monte Carlo sets for different fixed-parameter scenarios.

E. Comparison with Other Identifiability Approaches

Both methods detect the same parameter pairs, as expected: the CI of a parameter subset equals the inverse of the smallest singular value of its column-normalized sensitivity sub-matrix [10], so this agreement cross-validates both analyses. However, the CI in [3] was computed on time-averaged sensitivities, in which parameters affecting a single output ($Y_{\mathrm{X/gln}}$, $Y_{\mathrm{gln/glu}}$, $Y_{\mathrm{lac/glc}}$, $Y_{\mathrm{X/lac}}$) are collinear by construction; accordingly, [3] reported CI = ∞ for the glutamine pair and CI = 5×10^6 for the lactate pair. The time-resolved ROSM separates both cases. For glutamine, (10) depends only on the net yield $Y_{\mathrm{gln/glu}}/Y_{\mathrm{X/glu}} - 1/Y_{\mathrm{X/gln}}$, so GS-mediated synthesis and consumption cannot be distinguished from extracellular data: a genuine structural degeneracy. For lactate, the switch gives each parameter a distinct temporal signature, so the pair is structurally identifiable, and its weakness appears only as a practical limitation caused by the few post-switch samples (low t-value of $Y_{\mathrm{X/lac}}$). Moreover, a single singular value at machine precision shows that fixing one parameter suffices, a hierarchy that pairwise CI values do not provide.

Beyond [3], our results can be contrasted with the CI framework of Gábor et al. [9], on which the threshold used in [3] is based. That framework extends the CI to higher-order parameter groups by combinatorial search, showed that parameter triplets can be strongly collinear even when no pair is, and found only 13 practically identifiable parameters in a large-scale CHO metabolic model. Whereas exhaustive enumeration of collinear groups becomes infeasible beyond a few dozen parameters [9], SVD-ROSM recovers the non-

identifiable direction and its weights from a single decomposition, while the subsequent t-value analysis plays the role of their practical identifiability step. Consequently, clone comparisons in [3] involving K_{glc}, m_{glc} or $Y_{X/lac}$ should be interpreted cautiously (extended discussion in Supplementary S1).

IV. CONCLUSIONS

The MCKM was successfully implemented in Python with CasADi/IDAS, and the simulator was validated against the reference data. SVD-ROSM revealed a structural rank deficiency in the 13-parameter model, caused by the dependency between $Y_{X/gln}$ and $Y_{gln/glu}$. The consistency of the Monte Carlo analysis confirms that this dependency is an intrinsic property of the model structure rather than an artifact of the nominal values. Fixing $Y_{gln/glu}$ restored structural identifiability ($\sigma_{12} \approx 5 \times 10^{-3}$), well above the numerical noise threshold.

Nevertheless, structural identifiability alone did not guarantee practical identifiability: the 12-parameter model failed the t-value criterion under 5% noise and sparse sampling. Additionally fixing K_{glc}, $Y_{X/lac}$ and m_{glc} yielded a 9-parameter model with $|t| \geq 2.03$, in which all free parameters have a distinguishable influence on the outputs. Structural identifiability depends on the model architecture, whereas practical identifiability depends on the data, and both must be assessed to avoid spurious inferences in clone characterization.

Compared with the pairwise CI used in [3], SVD-ROSM evaluates the entire parameter space in a single decomposition, identifies the non-identifiable parameter combination, and distinguishes genuine structural dependencies from artifacts of time-averaged sensitivities. The open-source Python framework is reproducible, scalable, and extensible, facilitating its adoption in academic and industrial CLD.

V. SUPPLEMENTARY DATA

The code, all metabolite profiles, and the extended comparison with CI-based approaches (Supplementary S1) are available at https://github.com/betsabe9/MCKM-Identifiability-Analysis-CHO-cells.git

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